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The Modified-Athletic Shoulder Test: Reliability and validity of a new on-field assessment tool

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ABSTRACT

Objectives: To assess the reliability and the validity of a modified version of the Athletic Shoulder Test (M-AST), using a handheld dynamometer instead of a force plate.
Design: Test-retest reliability and concordance between tools.
Setting: University Motion Lab.
Participants: Twenty healthy volunteers who practiced an upper limb sport for at least 5 h per week were recruited for the study.
Main outcome measures: The concordance between the Athletic Shoulder Test (AST) and the Modified-Athletic Shoulder Test (M-AST) was assessed with intra-class correlation coefficients (ICC), paired T-Test and Bland-Altman plots. The inter-session reliability was assessed with intra-class correlation coefficients (ICC), SEM, MDC and paired T-Test.
Results: A strong concordance was found between AST and M-AST values (ICC = 0.86–0.97; $p > 0.05$) in all the positions considered and for both sessions. Bland-Altman plots confirmed these results. However, the inter-session reliability was more variable for both AST and M-AST (ICC = 0.643–0.923; $p < 0.05$).
Conclusions: The M-AST seems to be a reliable, cheaper and easier to implement alternative to the AST to measure recovery status following matches or training sessions in upper limb athletes. As for the AST, familiarization trials will have to be performed before the assessment to have accurate measurements.
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1. Introduction

Nowadays, the isokinetic device remains the gold standard in muscle strength assessment (Ellenbecker & Davies, 2000), (Forthomme, 2005). Indeed, this tool is considered as the most reliable, valid and safe means to assess muscle peak torque and evaluates muscle strength both in concentric and eccentric modes (Ellenbecker & Davies, 2000), (Forthomme et al., 2011). Whether at high or lower level of practice, shoulder rotator strength is often assessed with an isokinetic device in overhead athletes to highlight imbalances or shoulder injury risk factors (Forthomme et al., 2013), (Forthomme et al., 2018), (Tooth et al., 2020a). After a shoulder surgery or injury, this type of evaluation can help to determine whether an athlete is ready to go back to training or competition or not (Cools et al., 2021), (Ellenbecker & Davies, 2000), (Forthomme, 2005). However, isokinetic devices are expensive, which make them affordable only to major sport medicine centers which limits regular assessments for non-elite athletes.

The use of a handheld dynamometer to appreciate muscle function has become more popular in the last years for its accessibility and its cost (Cools et al., 2016), (Cools et al., 2015). Indeed, even if it still has a cost, this tool is more affordable to clinicians or fitness trainers and permits regular evaluations during the season. However, it remains less accurate and farther from sports gestures than the isokinetic dynamometer since only isometric strength can be measured. Even if it does not replace the isokinetic assessment, this tool can be considered as a complementary tool that permits a regular follow-up of an athlete.

Until now, the reliability as well as the validity of shoulder rotator strength measures done with a handheld dynamometer have been widely proven in literature (Cools et al., 2014), (Cools et al.,

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2016), (Popchak et al., 2021). However, the assessment of shoulder strength in non-rotational plane of movements has been less explored even if it can influence performance (Baena-Raya et al., 2021), (Baiget et al., 2022). Recently, Ashworth et al. (Ashworth et al., 2018) developed a new functional test (the Athletic Shoulder Test), aimed to assess shoulder strength in an antero-posterior plane, which was quite different from the evaluations suggested before. The positions considered for the test are close to some specific sport gestures such as the impact of the ball in tennis serve, the block in volleyball or the entry in the water in swimming. In practice, this test could be useful to assess residual or accumulated upper limb fatigue following training sessions or matches. However, the main limitation of the AST remains that it has been performed in a laboratory, with force plates, which makes it not affordable and inaccessible to non-elite athletes in practice.

Therefore, the aim of this study was to assess the reliability of a modified version of the Athletic Shoulder Test, using handheld dynamometer instead a force plate. Contrary to the Ashworth et al. (Ashworth et al., 2018) version, this modified version would permit regular on-field assessment of the athlete. Moreover, as it was not performed in Ashworth et al. (Ashworth et al., 2018) study, the validity of the test was also investigated in this study, through EMG measurements. Since scapular muscles are highly activated to protract, upwardly rotate and stabilize scapula in early pull through of freestyle swimming or at ball impact in volleyball and tennis, their activity will be explored during the test (Miura et al., 2020), (Pink et al., 1991), (Ryu et al., 1988).

2. Methods

2.1. Participants

Twenty male volunteers were enrolled in the study. They were aged 22.1 ± 2.1 years old, measuring 182.9 ± 5.4 m and weighing 82.6 ± 12.9 kilo's. Eighteen of them were right-handed and the two others were left-handed. They practiced various sports that were particularly soliciting for upper-limbs (tennis, volleyball, badminton, handball, crossfit etc …) with an average time of 8.3 ± 5.5 h/week for at least three years. Exclusion criteria were asymmetry of lower-limb length of, scoliosis or dorsal hyper-kyphosis, any shoulder surgical history, any shoulder pain, any shoulder injury (muscular, osseous, ligamentary, tendinous) or positive tendinous and impingement tests (Jobe, Patte 0°, Patte 90°, Lift off Test, Palm-up Test). Shoulder flexion range of motion was also measured in the sagital plane prior to inclusion to make sure that each participants had a complete range, without any trunk or lumbar compensations. Based on a confidence level of 95%, an expected reliability (ICC) of 0.9 and a precision of 0.1, the sample size had to be at least equal to 15 subjects (Bonett, 2002).

The protocol has been validated by the Medical Ethics Committee of the University Hospital of Liège. All the participants were informed about the nature of the tests and the progress of the experimentations before the beginning of the tests.

2.2. Procedure

The tests were organized into two different sessions, separated by 7 days in order to limit the influence of training on the results. The subjects were always assessed at the same moment of the day to limit the influence of external factors on the results. The Athletic Shoulder Test (AST) and the Modified-Athletic Shoulder Test (M-AST) were performed during both the first and the second sessions to evaluate the reliability and the concordance between the two tests. The AST and the M-AST were performed in a randomized order. During the first session, the specificity of the test was also evaluated by measuring muscle activity of scapular stabilizer muscles during this session. The two tests were performed on the dominant side in one half of the population and on the non-dominant side in the other half (in a randomized manner) to limit the influence of the dominance on the results. Unlike Ashworth et al., no familiarization trials were performed during the days before the tests in order to determine the usefulness (or not) of familiarization.

2.3. Warm-up

As a warm-up, the subjects were asked to perform: 20 internal and external rotations with an elastic band (green thera-band®, Akron, Ohio, USA), arm on their side and in a cocking position. Then, the participants performed 20 circumductions. Finally, they did two series of 10 overhead medicine ball throws. One minute was provided between the two series of medicine ball throws to avoid fatigue.

2.4. The Athletic Shoulder Test (AST)

The Athletic Shoulder Test has been described by Ashworth et al., in 2018 (Ashworth et al., 2018). To perform it, the participant is positioned prone on the floor, with their neck in a neutral position with a 4 cm foam block on which their forehead is rested. The forearm is in pronation, the elbow fully extended and the hand is placed on a force plate (Kistler ™ type 9281 EA, Kistler AG, Switzerland). The contralateral arm is placed behind the back to avoid compensations. After explaining the objective and the realization of the test, the participant's shoulder was first positioned in a I position (180° of abduction). A 5 cm-wood block was placed below the hand (corresponding to the height of a MicroFet 2, Hoggan Health Industries, Salt Lake City, USA) so that the arm was placed in the same way for the AST and the M-AST. In this position, the participant was asked to push as hard as possible on the force plate during 3 s. Three increasing sub-maximum trials of 6 s were done to get used to the exercice. Then, three maximal repetitions were done. Twenty seconds of rest were provided between maximal repetitions to avoid fatigue, as recommended by Ashworth et al. (Ashworth et al., 2018). Then, the participant's shoulder was positioned in a Y position (135° of abduction) and, finally, in a T position (90° of abduction) and the maximal contractions were performed with the same method. Only the best peak force value for each position was retained for analysis. As described by Ashworth et al. (Ashworth et al., 2018), the order of the position was always the same for all participants (I, Y and then T) (Fig. 1).

2.5. The Modified-Athletic Shoulder Test (M-AST)

The Modified-Athletic Shoulder Test was performed in the same position as the Athletic Shoulder Test. The participant was in a prone lying position on the floor, with their neck in a neutral position, their forehead on a 4 cm foam block and their contralateral arm on their back. Their hand was no longer placed on a force plate but on a handheld dynamometer with a height of 5 cm (MicroFet2: Hoggan Health Industries, Salt Lake City, USA). The center of the dynamometer was always placed 1 cm distally to the processus styloideus ulnae, at the level of the first carpal row. Tape was placed on their hand to make sure that the hand was positioned in exactly the same way for all the repetitions and for the whole sample. In the three positions considered (I, Y and T), the participant was asked to push as hard as possible on the handheld dynamometer during 3 s. Three maximal trials were performed after three sub-maximal trials. Twenty seconds of rest were provided between trials again. Only the best peak force value in each position was

Fig. 1. Illustration of the test ("Y position").

retained for analysis. The order of the positions was always the same for all the participants (I, Y and then T position) (Fig. 1).

2.6. EMG acquisitions

The electromyographic activity of three scapular muscles was investigated during the AST and the M-AST: the upper trapezius, the lower trapezius and the serratus anterior. The EMG signals were collected with Trigno Standard and Trigno Mini wireless sensors (Delsys, Boston, MA, USA) using silver-contact bipolar bar electrodes with fixed 10 mm inter-electrode spacing (Schwartz et al., 2017). After cleaning the skin with alcohol, the EMG electrodes were placed on it following Barbero et al. (Barbero et al., 2012) recommendations. The electrodes were only placed on the tested side. Data was acquired at a sample frequency of 1000 Hz.

Maximum voluntary isometric contractions were done at the end of the first session to normalize the EMG signal and to express it as a percentage of maximal activation. Maximum voluntary isometric contractions were performed in six differents positions as recommended by Schwartz et al. (Schwartz et al., 2017) and used in the Tooth et al. (Tooth et al., 2020b) study: the Empty Can, the Seated U 90°, the Seated U 125°, the Seated T, the Rotation 90° and the Prone-v-thumbs up (Fig. 2). To limit a possible influence of the investigator, the MVIC positions were maintained using a steel structure rather than manual pressure. This structure could adapt to both the test positions and the specific size of the volunteers. The volunteers were asked to perform three increasing sub-maximum trials of 6 s to get used to the exercice and the effort prior to each test position. Then, three maximum trials of 6 s were performed in each test position. All the tests were carried out bilaterally to increase muscle activation (Fischer et al., 2011) and to limit thoracic compensations. During the different trials, the volunteers were given verbal encouragement. To avoid fatigue, a minimum of 30 s rest intervals was provided between each trial and each MVIC test position respectively. This timing was chosen based on Schwartz et al. (Schwartz et al., 2017).

2.7. Data processing

The maximal isometric strength values recorded by the force plate or the handheld dynamometer were multiplied by upper limb length (measured between the acromion and the processus styloideus ulnae) to obtain the moment of force (expressed in Newton by meters).

The EMG signals were first band pass filtered (20–500 Hz, zero-phase 4th order Butterworth filter) and then filtered using a root-mean-square filter (500 ms moving window) (Schwartz et al., 2017). The EMG signals were then normalized with respect to the values obtained during the maximum voluntary contraction tests in order to obtain a percentage of the maximal voluntary activity (MVA) for each of the muscles considered. The maximal level of activation of each muscle (upper trapezius, lower trapezius and serratus anterior) was considered for analysis for all the subjects in the three positions (I, Y and T) for both AST and M-AST.

2.8. Statistical analysis

Statistical analysis was performed using SPSS Statistics (IBM, Armonk, New-York, USA). Descriptive statistics were calculated for all the conditions of testing. The normality of the variables was assessed with a Shapiro-Wilk test. All the variables were normally distributed. The relative reliability between the two tests (AST and M-AST) was assessed with an intraclass correlation coefficient (ICC 2,1). ICC values inferior to 0.5 indicated poor reliability, values between 0.5 and 0.75 indicated moderate reliability, values between 0.75 and 0.9 indicated good reliability and values greater than 0.9 indicated excellent reliability (Koo & Li, 2016). The absolute reliability between the two methods was explored with a paired-T test and using Bland-Altman plots. The inter-session relative reliability of the AST (force plate) and of the M-AST (handheld dynamometer) were separately determined using an intraclass correlation coefficient (ICC 2,1) as well as the standard error of measurement (SEM) as well as the minimal detectable change at 95% of confidence (MDC95) with the same method as Declève et al. (Declève et al.,

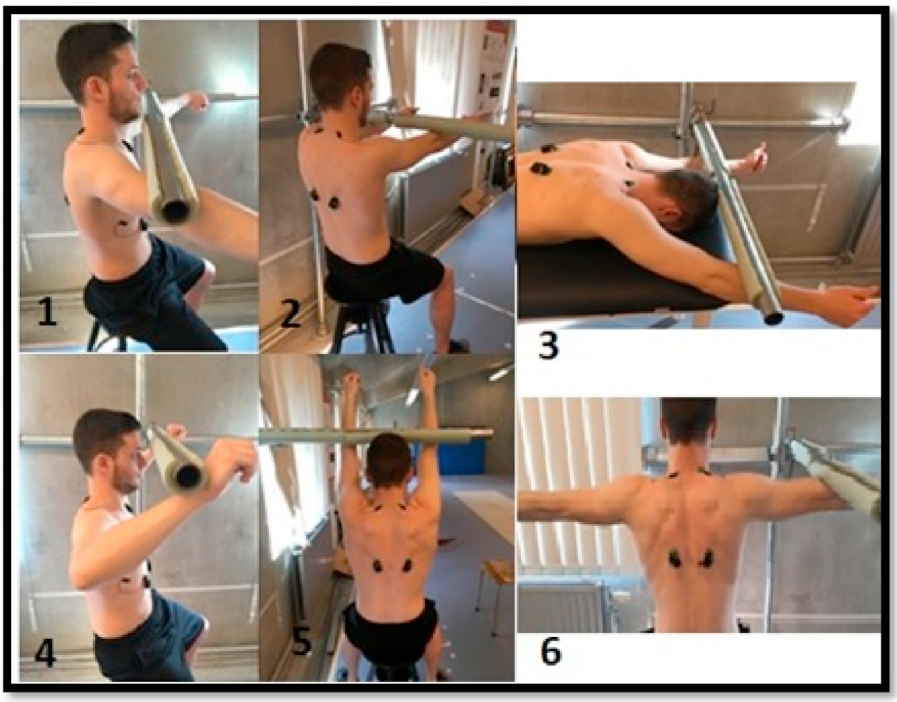

Fig. 2. Illustration of the Maximum Voluntary Isometric Contraction tests: (1) empty can, (2) seated U 90°, (3) prone-v-thumbs up, (4) rotation 90°, (5) seated U 125°, (6) seated T.

2021). The absolute inter-session reliability was assessed using a paired-T test. The ICC values were interpreted in the following way: ICC values less than 0.5 are indicative of poor reliability, values between 0.5 and 0.75 indicate moderate reliability, values between 0.75 and 0.9 indicate good reliability, and values greater than 0.90 indicate excellent reliability. Finally, muscle activity and maximal isometric strength were compared between the three positions (I, Y and T) using a one-way ANOVA. In case of significant p-values, Bonferroni method was used to compare the differences between pairs. The level of significance was set at $p < 0.05$ for all the tests performed.

3. Results

None of the participants reported pain before, during or after the tests, both for the first and the second sessions.

3.1. Descriptive statistics

Descriptive statistics of the two sessions are described in Table 1. For the first session, a significant difference in the isometric peak strength developed was found between the three positions of test both for the AST ($p = 0.002$) and the M-AST ($p = 0.018$). Indeed, the peak force developed in the I position was significantly more important than the one developed in the T position (0.021-0.002). This difference was still observed for the second session but did not reached significance ($p > 0.05$)

Table 1
Descriptive statistics (in N.m) of the test for the first and the second sessions for the two tests considered (AST and M-AST) [mean ± standard deviation].

	Test 1		Test 2	
	M-AST (in N.m)	AST (in N.m)	M-AST (in N.m)	AST (in N.m)
I position	89.28 ± 23.27	89.51 ± 20.61	92.01 ± 26.05	95.64 ± 27.07
Y position	77.03 ± 15.21	77.68 ± 14.17	85.85 ± 22.40	86.93 ± 20.29
T position	73.38 ± 13.97	71.02 ± 12.76	77.19 ± 16.85	80.70 ± 17.88
p-value	0.018[a]	0.002[a]	0.112	0.109

a = significant difference between I and T positions for post-hoc; b = significant difference between Y and T positions for post-hoc; c = significant difference between I and Y positions for post-hoc.

3.2. Concordance between AST and M-AST

The concordance between the AST (performed with a force plate) and the M-AST (performed with a handheld dynamometer) was assessed in three different ways: with an intra-class correlation coefficient (ICC), a paired T-test and with a Bland-Altman plot. The results of the different tests are mentionned in Table 2 while the Bland-Altman plots are presented in Appendix 1.

In general, the concordance between the two tests (and assessment methods) was considered as good to excellent (Koo & Li, 2016), with an ICC included between 0.864 and 0.952 and a p-value superior to 0.05, which significates that there is no significant difference between the tools for both sessions. The bias measured with the Bland-Altman method was also close to zero for all the positions considered. Finally, comparing the three positions, the "I position" seems to be the most reliable one while the "T position" can be considered as the less reliable of the three positions.

3.3. Inter-session reliability

The inter-session reliability was assessed in two different ways: with an intra-class correlation coefficient (ICC) and with a paired T-

Table 2
Concordance between the two tests (AST and M-AST) [ICC, p-value Student T-test, Bland-Altman bias and limit of agreement (LA)].

	ICC	T-Test(p)	Bias [LA 95%]
I position	0.952 [0.884–0.981]	0.864	−0.23 N m [-12.84-12.39]
Y position	0.926 [0.582–0.970]	0.611	−0.65 N m [-11.47-10.17]
T position	0.864 [0.688–0.944]	0.115	2.36 N m [-10.39-15.11]

test. The results of the different tests are mentionned in Table 3.

In general, the maximal isometric strength developped during the second session was higher than the one developped during the first session. This difference reached significance for all the positions using the force plate ($p < 0.05$) but only for the "Y position" with the handheld dynamometer measurements ($p = 0.011$). Therefore, we can consider that the absolute inter-session reliability is quite poor for both the AST and the M-AST. However, the relative reproducibility was good to excellent for the "I position" (ICC = 0.810–0.923) but moderate for the "Y position" (ICC = 0.643–0.743) and the "T position" (ICC = 0.715–0.826) (Koo & Li, 2016). If we consider the SEM and the MDC95 associated with the M-AST, we can observe that the MDC95 corresponds to 20.7% of the mean in the "I position" and 23.6% of the mean in the "T position", which is acceptable but it reached 39.49%" of the mean in the "Y position", which is quite high.

3.4. Scapular muscles activation

The mean activation of scapular muscles in the three positions of the test is summarized in Table 4. An important variability was found for muscle activation between subjects since the standard deviation is very high for all of the muscles and positions considered. Upper trapezius and lower trapezius seem to be at their highest activation level in the "I position" while the highest activation was found in the "T position" for the serratus anterior. Differences in activation between the three positions reached significance for the serratus anterior ($p = 0.025$) but not for the two other muscles ($p > 0.05$].

4. Discussion

McCall et al. (McCall et al., 2015) developed and validated an isometric strength evaluation of posterior limb muscles aimed to assess match-(or training) induced fatigue in football players. Inspired by these researchers, Ashworth et al. (Ashworth et al., 2018) suggested a similar test (the Athletic Shoulder Test) aimed to track residual or accumulated upper limb fatigue due to repetitive tackling movements in rugby players. However, although interesting in the follow-up of an athlete, the Athletic Shoulder Test (Ashworth et al., 2018) has to be performed with a force plate, which makes this test only affordable to a restricted percentage of athletes or clinicians. Therefore, the aim of this study was to assess the reliability a Modified version of the Athletic Shoulder Test (M-AST) using a handheld dynamometer instead of a force plate. The validity of this test was also explored by measuring scapular muscle activity during its.

Table 3
Inter-session reliability for the AST (force plate) and M-AST (MicroFet) [intra-class correlation coefficient (ICC), standard error of measurement (SEM), minimal detectable change with 95% of agreement (MDC95), percentage of the mean corresponding to the minimal detectable change (%mean) and p-value Student T-test].

		M-AST	AST
I position	ICC	0.923 [0.817–0.968]	0.810 [0.559–0.922]
	SEM	6.78 N m	10.44 N m
	MDC95(%mean)	18.78 N m (20.72%)	28.93 N m (31.25%)
	T-Test (p)	0.188	0.047
Y position	ICC	0.643 [0.233–0.849]	0.743 [0.096–0.917]
	SEM	11.60 N m	9.07 N m
	MDC95 (%mean)	32.17 N m (39.49%)	25.15 N m (30.56%)
	T-Test (p)	0.011	0.001
T position	ICC	0.826 [0.597–0.929]	0.715 [0.107–0.897]
	SEM	6.42 N m	8.59 N m
	MDC95 (%mean)	17.80 N m (23.65%)	23.82 N m (31.40%)
	T-Test (p)	0.059	0.002

Table 4
Upper trapezius, lower trapezius and serratus anterior activity (in % of Maximal Voluntary Activation) in the three positions of the test [mean ± standard deviation].

	Upper Trapezius	Lower Trapezius	Serratus Anterior
I position	27.8 ± 29.5	37.1 ± 40.3	27.5 ± 27.0
Y position	18.5 ± 19.7	19.8 ± 15.7	38.7 ± 22.4
T position	16.4 ± 13.1	22.8 ± 22.2	49.6 ± 24.9
p-value	0,223	0126	0,025[a]

The current study found a very good concordance between the AST and the M-AST, with an ICC above 0.9 for almost all the positions. The inter-session reliability was considered as moderate to good, no matter the measurement tool considered (force plate or handheld dynamometer). The reliability of force plates has been widely explored in the literature (Ashworth et al., 2018), (Verdera et al., 1999) as well as handheld dynamometers (Cools et al., 2016), (Popchak et al., 2021). The changes observed can, therefore, not be attributed to measurement errors. Two explanations can be advanced for this observation.

Firstly, we can see that the isometric peak strength was slightly increased in all the positions considered between the first and the second session (between 4 and 10%), which means that there may be a familiarization between the first and the second session. Increasing the time between the two sessions (for example 2 or 3 weeks) could have limited the learning effect between sessions but would probably have increased the risk of influence of external factors such as training. In Ashworth et al. (Ashworth et al., 2018) study, the participants performed three complete evaluations before the real evaluation, the participants in our study only performed sub-maximal trials before the evaluation. This choice was precisely done to determine the usefulness (or not) of a familiarization before the assessment. According to the results of the current study, a familiarization would be recommended before performing the test with athletes. However, in different clinical situations (such as an assessment after an injury or surgery), it is not always possible to perform familiarization sessions before the measurements. Therefore, in those cases, the values measured in practice will have to be interpreted with caution.

Secondly, the variability between the first and the second session can also be explained by the activities performed the days prior to the evaluation. Contrary to Ashworth et al. (Ashworth et al., 2018), we decided to let the subjects practice their sport normally in the days leading up to the assessment to be as close as possible to the reality of the field. Even if all of the subjects were assessed at the same moment of the week and the day and were asked to keep the same volume of activities during the weeks before and after the two measurements, some subjects developed a bit less strength during the second session because of the intensity of the match or the training session they had the previous days. This strengthens the interest of the AST or the M-AST in assessing recovery status after training sessions or match.

Concerning the activity of scapular muscles, we observed that for the upper trapezius, the highest activation was in the "I position" and the lowest one in the "T position". The opposite was noticed for the serratus anterior, with the highest activation in the "T position" and the lowest one in the "I position". As for the lower trapezius, the highest activation was measured in the "I position", followed by the "T position" and then the "Y position".

Except for the serratus anterior which was most activated in the "T position", no significant difference in activation was measured between positions for the upper or the lower trapezius. The activation values observed for the upper and lower trapezius in the current study were also inferior to the ones reported in literature for tennis serve or volleyball spikes (Escamilla & Andrews, 2009),

(Miura et al., 2020), (Ryu et al., 1988). This can be explained by the fact that the participants were performing isometric contractions in a supine position. Therefore, the upper trapezius muscle did not have to elevate the arm against gravity and the scapular stabilizers did not have to actively contribute to scapular upward rotation because of the position of the test.

On the other hand, the serratus anterior activation recorded in the current study appeared to be close to the values measured at late recovery or early pull-through in freestyle swimming (30–33%) (corresponding to hand entry) or at ball impact in volleyball spike (42–43%) (Miura et al., 2020), (Pink et al., 1991). As for the difference in serratus anterior activation observed between the positions, this can be related to the findings of Escamilla et al. (Escamilla & Andrews, 2009). Indeed, in tennis, these authors reported a higher serratus anterior activation in a forehand motion (corresponding to the "T position") than in the serve motion (corresponding to the "I" or "Y" position), which is quite similar to our findings.

However, the standard deviations were quite high and almost equal to the mean activation values (especially for the upper and lower trapezius), which means that variability between subjects is quite high. Therefore, we can suppose that different muscular strategies were used by the subjects to develop strength in the three positions considered in the test. In the current study, muscle activity was only measured during the first session, when the participants were less familiarized with the test. In the future, scapular stabilizer activity should be measured in subjects that are well familiarized with the test to observe if the variability is still very important.

In practice, the Modified-Athletic shoulder test (M-AST) appears to be a reliable alternative to measure maximal isometric strength in the follow-up of upper limb athletes. Contrary to the test described by Ashworth et al. (Ashworth et al., 2018), the use of a handheld dynamometer instead of a force plate makes it more affordable to clinicians, trainers and all kind of athletes even if this tool still has a cost. Moreover, positioning the handheld dynamometer on the ground permits a good stabilization of the device and reduces the influence of experimenter's strength on the results (Kolber & Cleland, 2013), (Wikholm & Bohannon, 1991). The best concordance between AST and M-AST was found in the "I position". Therefore, if a single position had to be kept (because of lack of time for example), it would be the "I position".

In overhead athletes, this test could be used for three different objectives. Firstly, the M-AST could be useful to guide performance while measuring isometric peak force in positions close to upper limb gestures such as ball impact in tennis, block in volleyball or the entry in water in swimming. Secondly, this test could be probably integrated in a battery of tests aimed to determine return to play capacity after an injury or surgery in upper limb athletes, even if future prospective studies will have to be performed to confirm these words. As the current study assessed "healthy" upper limb athletes, the data provided here can be considered as "healthy" normative data and used as such in practice. However, the included participants performed different overhead sports and results mixed dominant and non-dominant arm values. Therefore, in the future, sport-specific normative data could be established for better specificity. Finally, as described in McCall et al. study (McCall et al., 2015), an isometric strength evaluation can provide indications of the residual or accumulated fatigue after training sessions or matches while decreasing the risk of instant injury (which is more frequent with concentric or eccentric contractions than with isometric ones). This evaluation could be complementary to training load calculations in the management of an athlete.

The handheld dynamometer is easily transportable, which permits regular assessments on the field. Since reductions of muscle force have been correlated to an increase in injury risk (Gabbett, 2016) in rugby players, the results of this test could give information to trainers to adapt training or to determine the capacity of an athlete to perform at 100% in competition. Since limb length influences the strength developed by the subject at the shoulder joint, it appeared essential, in the current study, to express the results as joint moments (in newton-meters), contrary to what had been done by Ashworth et al. (Ashworth et al., 2018). Indeed, in practice, only considering absolute values (in newtons) does not allow comparisons between subjects or to normative data.

This study has limitations. The main one is the variability of the sport practiced by the participants. Even if they were all playing upper limb sports, some of them were practicing handball, volley or tennis while others practiced kayaking or boxing. The number of hours of practice was also different between subjects, which may have influenced muscle activation during the isometric contractions. These factors do not have an influence on the validity and reliability of the test but condition the normative data established.

Another limitation has to be highlighted. Although more affordable than a force platform, a handheld dynamometer still has a cost. Nowadays, a lot of therapists have access to this tool in their practice (the proposed model or a cheaper one) but it is not the case for everyone. Since only peak force is measured in this protocol, we could imagine reproducing the protocol with a weight scale for those who do not have access to such materials. A last limitation concerns the choice of the muscles for EMG measurements. Pectoralis major and subscapularis have both been demonstrated as highly activated in the sports gestures considered (entry in water in swimming, ball impact in tennis/volleyball) and could have been explored too. However, subscapularis activity is challenging to measure with surface electrodes since it is a deep muscle. Then, pectoralis major activity measurement would have induced a discomfort for the participants since they would have been lying on the electrodes. This could have influenced the measurements too.

5. Conclusion

The Modified-Athletic Shoulder Test (M-AST) constitutes a reliable, quick and easy to implement test to measure performance and return to play capacity in a non-rotational plane of movements in overhead athletes. The recovery status after training sessions/matches could also be monitored with this test. Among the 3 positions considered for the test, the "I position" seems to be the most reliable one.

The use of a handheld dynamometer instead of a force plate makes it more affordable for clinicians, trainers and all kinds of athletes. Since only isometric contractions are performed, the risk of injury related to its realization is very low. However, a familiarization with the test is probably necessary before performing it to ensure accurate measurements.

Future studies will be necessary to further explore the usefulness of this test in practice, its validity and the impact of training sessions or competitions on the peak forces measured. Normative sport-specific values will have to be established for this test too.

Ethical statement

The protocol has been validated by the Medical Ethic Committee of the Liège University Hospital (file number 2021-277). All the participants were informed about the nature of the experimentations and signed a consent form.

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Declaration of competing interest

The authors, their immediate families, and any research foundation with which they are affiliated did not receive any financial payments or other benefits from any commercial entity related to the subject of this article.

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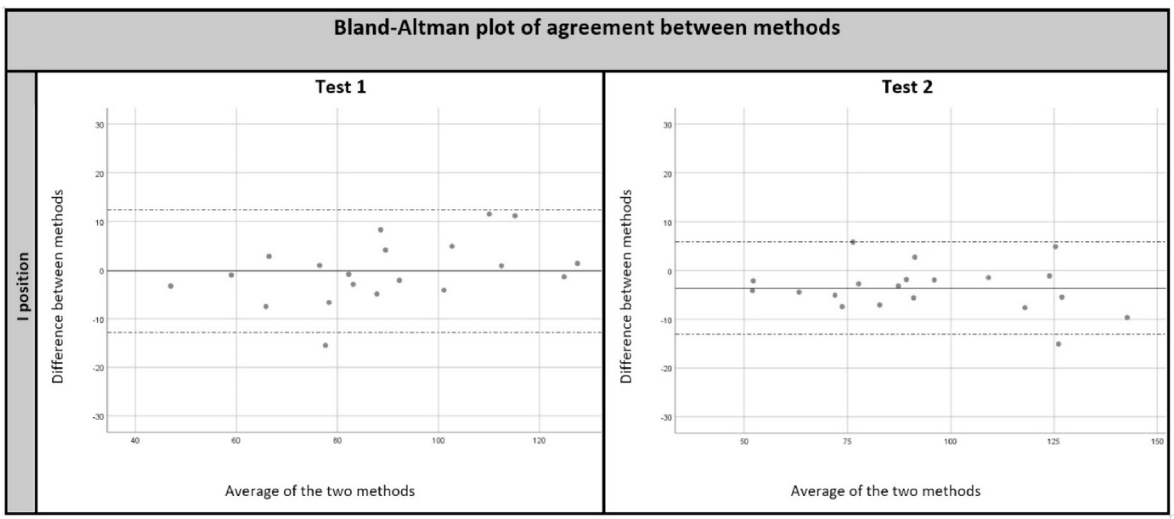
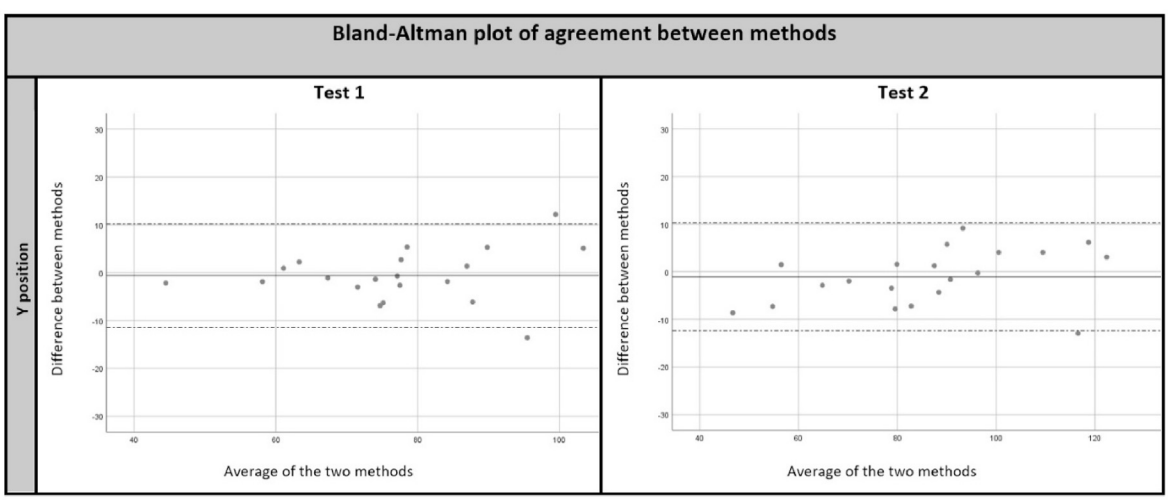
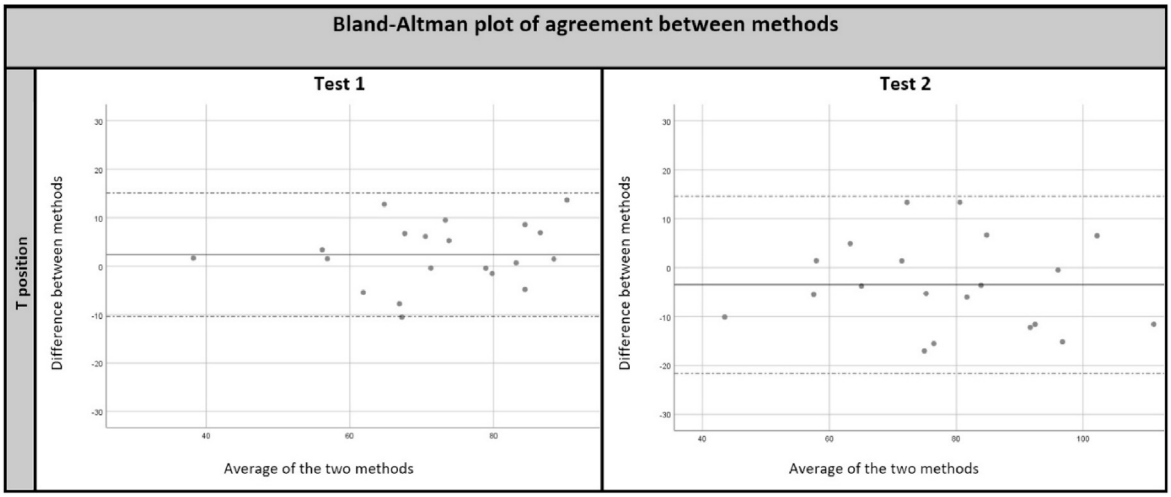
Appendix 1



Bland-Altman plot of agreement between methods for “I position”



Bland-Altman plot of agreement between methods for “Y position”



Bland-Altman plot of agreement between methods for “T position”

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